

1 Cylindrical algebraic decomposition

Theorem 1.1 Thom lemma. Let $f_1, \dots, f_k \in \mathbb{R}[T]$ be nonzero polynomials such that if $f'_i \neq 0$, then $f'_i \in \{f_1, \dots, f_k\}$. Let $\epsilon : \{1, \dots, k\} \rightarrow \{-1, 0, 1\}$, and put

$$A_\epsilon := \{t \in \mathbb{R} : \text{sign}(f_i(t)) = \epsilon(i), i = 1, \dots, k\}, \text{ a subset of } \mathbb{R}.$$

Then A_ϵ is empty, a point, or an interval.

Moreover, If $A_\epsilon \neq \emptyset$, then its closure is given by

$$\text{cl}(A_\epsilon) = \{t \in \mathbb{R} : \text{sign}(f_i(t)) \in \{\epsilon(i), 0\}, i = 1, \dots, k\}.$$

If $A_\epsilon = \emptyset$, then $\text{cl}(A_\epsilon)$ is empty or points.

Proof. By induction on s . The Lemma holds trivially for $s = 0$. Let $f_1, \dots, f_s, f_{s+1} \in R[\mathbf{x}] \setminus \{0\}$ be polynomials such that if $f'_k \neq 0$, then $f'_k \in \{f_1, \dots, f_{s+1}\}$. Without loss of generality we assume that $\deg(f_{s+1}) = \max\{\deg(f_k) : 1 \leq k \leq s+1\}$.

Let $\epsilon' : \{1, \dots, s, s+1\} \rightarrow \{-1, 0, 1\}$ and $\epsilon : \{1, \dots, s\} \rightarrow \{-1, 0, 1\}$ the restriction.

Note that

$$A_{\epsilon'} = A_\epsilon \cap \{x \in R : \text{sign}(f_{s+1}(x)) = \epsilon'(s+1)\}.$$

By induction A_ϵ is empty, a point, or an interval.

If A_ϵ is empty or a point, then obviously so is $A_{\epsilon'}$ and the other property follows immediately by induction hypothesis on A_ϵ .

Assume A_ϵ is an interval. Now $f'_{s+1} = 0$ or $f'_{s+1} \in \{f_1, \dots, f_s\}$. So by definition of A_ϵ , f'_{s+1} has constant sign on A_ϵ . Therefore f_{s+1} is either strictly increasing, or strictly decreasing or constant on A_ϵ .

Consider $A_\epsilon = (a, b)$. There are three cases depending on $\epsilon'(s+1)$:

Case 1. $A_{\epsilon'} = \{x \in (a, b) : f_{s+1}(x) > 0\}$.

Case 2. $A_{\epsilon'} = \{x \in (a, b) : f_{s+1}(x) < 0\}$.

Case 3. $A_{\epsilon'} = \{x \in (a, b) : f_{s+1}(x) = 0\}$.

If $A_{\epsilon'} = \emptyset$ there is nothing to prove.

Assume $A_{\epsilon'} \neq \emptyset$. If f_{s+1} is constant on A_ϵ then f_{s+1} is a constant polynomial $f_{s+1}(x) = c \neq 0$. So $A_{\epsilon'}$ is empty or $A_{\epsilon'} = (a, b)$ depending on whether $\text{sign}(c) = \epsilon'(s+1)$.

Assume now f_{s+1} strictly increasing on A_ϵ and $A_{\epsilon'} = \{x \in (a, b) : f_{s+1}(x) > 0\} \neq \emptyset$. Let $x_0 = \inf\{x \in (a, b) : f_{s+1}(x) > 0\}$. Since f_{s+1} is strictly increasing it follows that $f_{s+1}(x) > 0 \forall x \in (a, b)$ with $x > x_0$. So $A_{\epsilon'} = (x_0, b)$ and its closure is $[x_0, b] = A_{\epsilon'}$. The other cases are treated similarly.

□

Notation. X is a nonempty topological space, E a ring of continuous real-valued functions $f : X \rightarrow \mathbb{R}$, the ring operations being pointwise addition and multiplication, with multiplicative identity the function on X that takes the constant value 1.

For $f(T)$ as above and $z \in X$, we write $f(z, T)$ for the polynomial

$$f(z, T) = f_0(z) + f_1(z)T + \dots + f_d(z)T^d \in \mathbb{R}[T].$$

Let us say that a polynomial $f(T) \in E[T]$ has *zero-free leading coefficient* if

$$f(T) = c_d T^d + c_{d-1} T^{d-1} + \cdots + c_0$$

with $c_i \in E$, such that c_d has no zero on X .

Definition 1.1. Call a set $A \subseteq X$ an E -set if A is a finite union of sets of the form

$$\{x \in X : f(x) = 0, g_1(x) > 0, \dots, g_k(x) > 0\}, \text{ with } f, g_1, \dots, g_k \in E.$$

Note that the E -sets form a boolean algebra of subsets of X . unions, intersections and taking complements.

Definition 1.2. If $X = \mathbb{R}^m$, and $E = \mathbb{R}[x_1, \dots, x_m]$ is the ring of real polynomial functions on X , then the E -sets are exactly the semialgebraic subsets of $\mathbb{R}^m$.

Lemma 1.1 Lemma 2. Let $G(A, T) = A_0 + A_1 T + \cdots + A_d T^d \in \mathbb{Z}[A, T]$, $A = (A_0, \dots, A_d)$, be a general polynomial of degree d , and $e \in \{0, \dots, d, \infty\}$. Then the set

$$\{a \in \mathbb{R}^{d+1} : G(a, T) \text{ has exactly } e \text{ distinct complex zeros}\}$$

is a finite union of sets of the form

$$\{a \in \mathbb{C}^{d+1} : p_1(a) = \cdots = p_k(a) = 0, q(a) \neq 0\},$$

where $p_i, q \in \mathbb{Z}[A]$.

Proof. The cases $d = 0$ or $e \in \{0, \infty\}$ are trivial. Let $d > 0, e \in \{1, \dots, d\}$, and $a = (a_0, \dots, a_d) \in \mathbb{C}^{d+1}, a_d \neq 0$.

Let $g = \text{degree of GCD}(G(a, T), \frac{\partial G}{\partial T}(a, T))$ in $\mathbb{C}[T]$.

Then the number of distinct complex zeros of $G(a, T)$ is $d - g$, and the degree of $\text{LCM}(G(a, T), \frac{\partial G}{\partial T}(a, T))$ is $2d - g - 1$.

Hence the condition is that $G(a, T)$ has at most e distinct zeros, which is equivalent to $d - g \leq e$, that is, to $2d - g - 1 \leq d + e - 1$.

The last condition is equivalent to the condition: (*) There exist $q(x, T) = x_0 + x_1 T + \cdots + x_{e-1} T^{e-1}$ and

$$r(x, T) = x_e + x_{e+1} T + \cdots + x_{2e} T^e,$$

with

$$x = (x_0, \dots, x_{2e}) \in \mathbb{C}^{2e+1} \setminus 0,$$

such that

$$G(a, T)q(x, T) = \frac{\partial G}{\partial T}(a, T)r(x, T).$$

This equality can be rewritten as

$$G(a, T)q(x, T) - \frac{\partial G}{\partial T}(a, T)r(x, T) = \beta_0(a, x) + \beta_1(a, x)T + \cdots + \beta_{d+e-1}(a, x)T^{d+e-1}$$

where $\beta = (\beta_0, \dots, \beta_{d+e-1}) : \mathbb{C}^{d+1} \times \mathbb{C}^{2e+1} \rightarrow \mathbb{C}^{d+e-1}$ is a bilinear function.

So $(*)$ is equivalent to the condition $\beta_0(a, x) = \cdots = \beta_{d+e-1}(a, x) = 0$ has nonzero solution $x \in \mathbb{C}^{2e+1}$. The last condition is equivalent to the vanishing of all $(2e+1)$ -minor of the matrix of the linear map $\beta(a, \cdot)$. Note that each of the minors is a polynomial in $a_0, \dots, a_d$ with coefficients in $\mathbb{Z}$. Therefore, for each $d' \leq d$, the set $M_e^{d'} = \{a \in \mathbb{R}^{d+1} : G(a, T) \text{ is of degree } d' \text{ and has at most } e \text{ distinct complex zeros}\}$ is the intersection of the set $\{a \in \mathbb{R}^{d+1} : a_d = \cdots = a_{d'+1} = 0, a_{d'} \neq 0\}$ with the zero set of certain polynomials in $\mathbb{Z}[A]$. So $\{a = (a_0, \dots, a_d) \in \mathbb{R}^{d+1} : G(a, T) \text{ has exactly } e \text{ complex zeros}\} = \bigcup_{d'=0}^d M_e^{d'} \setminus M_{e-1}^{d'}$ a semi-algebraic set. $\square$

Corollar 1.1. As a consequence, for every $f \in \mathbb{R}[X_1, \dots, X_n, T] = \mathbb{R}[X][T]$,

$$f(X_1, \dots, X_n, T) = a_0(X) + a_1(X)T + \cdots + a_d(X)T^d,$$

the set

$$\{x \in \mathbb{R}^n : f(x, T) \text{ has exactly } e \text{ distinct complex zeros}\}$$

is a semi-algebraic subset of $\mathbb{R}^n$.

Proof. Since $f(x, T) = G(a_0(x), \dots, a_d(x), T)$, the corollary follows from Lemma 2. $\square$

Theorem 1.2 Cylindrical algebraic decomposition. Let $f_1(T), \dots, f_M(T) \in E[T]$. Then the list $f_1, \dots, f_M$ can be augmented to a list $f_1, \dots, f_N$ in $E[T]$ ($M \leq N$), and X can be partitioned into finitely many E -sets X_i ($1 \leq i \leq k$) such that for each connected component C of each X_i there are continuous real-valued functions $\xi_{C,1} < \cdots < \xi_{C,m(C)}$ on C with the following two properties:

- i) Each function f_n ($1 \leq n \leq N$) has constant sign ($-1, 0$, or 1) on each of the graphs $\Gamma(\xi_{C,j})$ ($1 \leq j \leq m(C)$) and on each of the sets $(\xi_{C,j}, \xi_{C,j+1})$ ($0 \leq j \leq m(C)$), where $\xi_{C,0} := -\infty$ and $\xi_{C,m(C)+1} := +\infty$ are constant functions on C by convention;
- ii) Each of the sets $\Gamma(\xi_{C,j})$ and $(\xi_{C,j}, \xi_{C,j+1})$ from (1) is of the form

$$\{(x, t) \in C \times \mathbb{R} : \text{sign}(f_n(x, t)) = \epsilon(n) \text{ for } n = 1, \dots, N\}$$

for a suitable sign condition $\epsilon : \{1, \dots, N\} \rightarrow \{-1, 0, 1\}$.

Proof. Let $d = \max\{\deg_T(f_i), i = 1, \dots, p\}$. Augment $f_1, \dots, f_p$ to

$$\{f_1, \dots, f_{p+q}\} = \left\{ \frac{\partial^\nu f_i}{\partial T^\nu} : 1 \leq i \leq p, 0 \leq \nu \leq d \right\}.$$

For each $\Delta \subset \{1, \dots, p\} \times \{0, \dots, d\}$, and $e \in \{0, \dots, pd^2\} \cup \{\infty\}$, put

$$f_\Delta(T) = \prod_{(i,\nu) \in \Delta} \frac{\partial^\nu f_i}{\partial T^\nu} \in \mathbb{R}[X][T], \text{ and}$$

$$A_{\Delta,e} = \{x \in \mathbb{R}^n : f_\Delta(x, T) \text{ has exactly } e \text{ complex zeros}\}.$$

By Lemma 2, $A_{\Delta,e}$ is a semi-algebraic set. For a given Δ the family $\{A_{\Delta,e} : e \text{ varies}\}$ forms a partition of $\mathbb{R}^n$. Since the class of semi-algebraic sets is a boolean algebra we can

find a partition (the intersection of the partitions) $\mathbb{R}^n = S_1 \cup \cdots \cup S_k$, where each S_i is semi-algebraic such that each set $A_{\Delta,e}$ is a union of the S'_i s. We will prove that $f_1, \dots, f_{p+q}$ and $S_1, \dots, S_k$ satisfy the conclusion of the theorem.

For each connected component C of S_i put

$$\Delta(C) = \left\{ (i, \nu) : \frac{\partial^\nu f_i}{\partial T^\nu} \neq 0 \text{ on } C \times \mathbb{R} \right\}.$$

By Lemma 3, there exist continuous functions $\xi_{C,1} < \cdots < \xi_{C,r}(C)$ on C such that $\{(x, t) \in C \times \mathbb{R} : f_\Delta(C) = 0\} = \Gamma(\xi_{C,1}) \cup \cdots \cup \Gamma(\xi_{C,r}(C))$.

Check (i): If $(i, \nu) \notin \Delta(C)$ then $\frac{\partial^\nu f_i}{\partial T^\nu} \equiv 0$ on the sets given in (i).

If $(i, \nu) \in \Delta(C)$, then $C \subset A_{\{(i,\nu)\},e}$, for certain $e \in \{0, \dots, d\} \cup \{\infty\}$ and the number of real zeros of $\frac{\partial^\nu f_i}{\partial T^\nu}(x, T)$ is independent of $x \in C$.

Since $\frac{\partial^\nu f_i}{\partial T^\nu}$ is a factor of $f_\Delta(C)$, by Lemma 3, the zeros of $\frac{\partial^\nu f_i}{\partial T^\nu}(x, T)$, for $x \in C$, must be among the $\xi_{C,j}(x)$'s. Since C is connected, (i) is checked.

Check (ii): Let B be one of the sets in (i). By (i), $\epsilon(i, \nu) = \text{sign}(\frac{\partial^\nu f_i}{\partial T^\nu}|_B)$ is well-defined. Put

$$B' = \left\{ (x, t) \in C \times \mathbb{R} : \text{sign} \left(\frac{\partial^\nu f_i}{\partial T^\nu}(x, t) \right) = \epsilon(i, \nu), 1 \leq i \leq p, 0 \leq \nu \leq d \right\}.$$

Clearly $B \subset B'$. If $B \neq B'$ then exist $(x, t') \in B' \setminus B$, $(x, t) \in B$ (say $t < t'$). Thom's lemma implies that $\{r \in \mathbb{R} : (x, r) \in B'\}$ is connected, so $\{x\} \times [t, t'] \subset B'$. Since $(x, t) \in B$, $(x, t') \notin B$, $f_\Delta(c)$ must change sign on $\{x\} \times [t, t']$. But $f_\Delta(c)$ is a product of $\frac{\partial^\nu f_i}{\partial T^\nu}$, so $f_\Delta(c)$ cannot change sign on B' , contradiction. Therefore $B = B'$. □

Theorem 1.3 Lojasiewicz. The number of connected components of a semi-algebraic set is finite, and each of the components is also semi-algebraic.

Corollar 1.2. If (X, E) has the Lojasiewicz property, then $(X \times \mathbb{R}, E(T))$ also has the Lojasiewicz property, and the image of any $E[T]$ -set $S \subseteq X \times \mathbb{R}$ under the projection map $X \times \mathbb{R} \rightarrow X$ is an E -set.

Proof. Write an $E[T]$ -set S as a finite union of sets of the form

$$\{(x, t) \in X \times \mathbb{R} : f(x, t) = 0, g_1(x, t) > 0, \dots, g_k(x, t) > 0\}$$

with $f, g_1, \dots, g_k \in E[T]$. Let $f_1, \dots, f_M$ be a list of all the polynomials in $E[T]$ that are involved in such a definition of S . Now apply the theorem to this list. □

Theorem 1.4 Tarski-Seidenberg Theorem. The $\mathbb{R}[T_1, \dots, T_m]$ -sets in the space $\mathbb{R}^m$ are exactly the semialgebraic subsets of $\mathbb{R}^m$. For $m = 0$ the space $\mathbb{R}^m$ has only one point, so the pair $(\mathbb{R}^m, \mathbb{R}[T_1, \dots, T_m])$ obviously has the Lojasiewicz property for $m = 0$; hence by induction on m , using (2.9), it follows that $(\mathbb{R}^m, \mathbb{R}[T_1, \dots, T_m])$ has the Lojasiewicz property for each m , and that the image of a semialgebraic subset of $\mathbb{R}^{m+1}$ under the projection map $\mathbb{R}^{m+1} \rightarrow \mathbb{R}^m$ is a semialgebraic subset of $\mathbb{R}^m$.

We can now draw the following conclusion about the model-theoretic structure $(\mathbb{R}, <, 0, 1, +, -, \cdot)$, which we will call “the ordered field of real numbers”.

Corollar 1.3. The sets $S \subseteq \mathbb{R}^m$ ($m = 0, 1, 2, \dots$) that are definable using constants in the ordered field of real numbers are exactly the semialgebraic sets. The ordered field of real numbers is o-minimal.

Proof. semialgebraic sets form a structure on $(\mathbb{R}, <)$. Clearly the primitives of the ordered field of reals are semialgebraic, the semialgebraic sets are definable using constants in the ordered field of real numbers. Finally, the semialgebraic subsets of $\mathbb{R}$ are clearly the finite unions of intervals and points. The desired result follows. □

2 noether normalization

2.1 noether normalization

Let $f(X_1, \dots, X_n)$ be a nonzero polynomial of degree d over a field K of characteristic 0, say

$$f = \sum_{|i|=d} c_i X^i + \text{terms of lower degree}, \quad n > 0.$$

Here $i = (i_1, \dots, i_n) \in \mathbb{N}^n$, $|i| = i_1 + \dots + i_n$ and $X^i = X_1^{i_1} \dots X_n^{i_n}$, $c_i \in K$. Let $(\lambda_1, \dots, \lambda_{n-1}) \in K^{n-1}$. We are going to make the (invertible) substitution

$$\begin{aligned} X_1 &\rightarrow X_1 + \lambda_1 X_n, \\ &\vdots \\ X_{n-1} &\rightarrow X_{n-1} + \lambda_{n-1} X_n, \\ X_n &\rightarrow X_n. \end{aligned}$$

Note that, with $\lambda^i = \lambda_1^{i_1} \dots \lambda_{n-1}^{i_{n-1}}$, this substitution transforms f into

$$\begin{aligned} &f(X_1 + \lambda_1 X_n, \dots, X_{n-1} + \lambda_{n-1} X_n, X_n) \\ &= \left(\sum_{|i|=d} c_i \lambda^i \right) X_n^d + \text{terms of degree } < d \text{ in } X_n. \end{aligned}$$

Since the polynomial $\sum_{|i|=d} c_i Y_1^{i_1} \dots Y_{n-1}^{i_{n-1}} \in K[Y_1, \dots, Y_{n-1}]$ is nonzero, it follows that for “almost all” $\lambda = (\lambda_1, \dots, \lambda_{n-1}) \in K^{n-1}$, namely for all those λ that are not zeros of this polynomial, the transformed polynomial

$$f(X_1 + \lambda_1 X_n, \dots, X_{n-1} + \lambda_{n-1} X_n, X_n)$$

is of (total) degree d , and has a term of the form cX_n^d with $c \in K - \{0\}$.